Week 04 teacher guide

Dividing fractions

PowerPoint slides 298-394 | Student workbook pp. 30-37

Division asks how many divisor-sized groups fit. Equal small units make the comparison visible. The reciprocal rule records that reasoning efficiently.

The seven days

Day 022 · p. 30 · Saturday · 20 questions,including 2 challenges
The meaning of fraction division

Day 023 · pp. 31-32 · Sunday · 20 questions,including 2 challenges
Fraction division and reciprocals

Day 024 · p. 33 · Monday · 20 questions,including 2 challenges
Dividing fractions

Day 025 · p. 34 · Tuesday · 20 questions,including 2 challenges
Dividing fractions

Day 026 · p. 35 · Wednesday · 25 questions,including 2 challenges
Dividing fractions

Day 027 · p. 36 · Thursday · 20 questions,including 2 challenges
Dividing fractions

Day 028 · p. 37 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 024: Fraction operations (9, 10, 11, 12)

Day 025: Fraction operations (9, 10, 11, 12)

Day 026: Weekly puzzle (13)

Day 027: Percent practice (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. The meaning of fraction division

Day 022 · p. 30 | Slides 298-326

Goal: I can show how many fractional groups fit in an amount.

Check first: Fraction equivalence and division by unit fractions.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 298-299 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 300-301 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 302-303 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 304 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 305-306 · Explore · P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 307-309 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 310 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 311 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 312 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 313 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 314-316 · Guided practice · X1, X2, X3

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 317-318 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 319-322 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 323 · Summary · S

Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 324 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 325 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 326 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: A remainder has two possible units

Watch and discuss; this is a projected investigation, not an extra assigned question.

Model $2\ \frac{1}{3} \div \frac{2}{3}$. Explain both the exact quotient and the number of full pieces. Identify the remainder as a length and as a fraction of one piece.

1. Rename the total as 7 third-meter cells.
2. Circle three full groups of 2 cells.
3. Name the leftover length as $\frac{1}{3}$ meter.
4. Compare the leftover cell with a 2-cell piece to explain the $\frac{1}{2}$ in the quotient.

Ask: What does the denominator 2 in the quotient measure, and how is that different from the denominator 3 in the leftover length?

If stuck: Put unit labels after every fraction: meters or pieces.

Expected: Exact quotient $\frac{7}{2} = 3\ \frac{1}{2}$ pieces; 3 full pieces; remainder $\frac{1}{3}$ meter, which is $\frac{1}{2}$ of a $\frac{2}{3}$-meter piece.

C1 · Challenge discussion

Ask: What unit does the fractional part of the quotient describe?

If stuck: Draw ten fourths and circle groups of three fourths.

Expected: 3 full pieces; $\frac{1}{4}$ meter remains; quotient $\frac{10}{3}$ counts 3 and $\frac{1}{3}$ of a piece

C2 · Challenge discussion

Ask: Why can two different quotients give the same number of full cups?

If stuck: Distinguish the exact number of cupfuls from the whole number of filled cups.

Expected: A: 2 cups; B: $\frac{12}{5}$ cups; both fill 2 full cups. B has $\frac{1}{6}$ liter left

What to notice

Labels what counts as one group. → Ask the student to circle one whole group.

Renames both amounts using the same small unit. → Accept a tape or a number line with equal units.

Common error: Reverses dividend and divisor. Repair: Label amount first and group size second.

Sunday. Fraction division and reciprocals

Day 023 · pp. 31-32 | Slides 327-354

Goal: I can connect the reciprocal rule to equal groups and use it in measurement problems.

Check first: Fraction division with equal-part models and mixed-number conversion.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 327-328 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 329-330 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 331-332 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 333 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 334-336 · Explore · P4, P5, P6, P7

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 337-339 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 340 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 341 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 342 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 343 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 344 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 345-346 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 347-350 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 351 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 352 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 353 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 354 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: A reciprocal counts groups in one whole

Watch and discuss; this is a projected investigation, not an extra assigned question.

Explain $\frac{3}{4} \div \frac{2}{5}$ in two ways: use twentieths to count equal pieces, and count how many $\frac{2}{5}$-sized groups are in one whole before taking three fourths of that count.

1. Rename dividend and divisor in twentieths.
2. Count 15 pieces in groups of 8.
3. Find groups per whole: $\frac{5}{2}$.
4. Take $\frac{3}{4}$ of that group count and connect it to the reciprocal rule.

Ask: Where does $\frac{5}{2}$ come from in the model? Why does it multiply the dividend?

If stuck: Ask how many 2-fifth groups fit in 5 fifths before considering $\frac{3}{4}$ of a whole.

Expected: $\frac{15}{8} = 1\ \frac{7}{8}$ groups. Both methods count the same groups.

C1 · Challenge discussion

Ask: Why does dividing area by width recover a length? Why is perimeter not another product?

If stuck: Label a rectangle with the known area and width before choosing operations.

Expected: Length $\frac{5}{4}$ meters; perimeter $\frac{23}{6}$ meters

C2 · Challenge discussion

Ask: Where do square units become cubic units in your calculation?

If stuck: Find the base area first, then multiply by height.

Expected: Each box: $\frac{5}{4}$ m³; unoccupied volume: $\frac{1}{2}$ m³

What to notice

Converts the mixed number before dividing. → Ask how many eighths make both whole yards.

Uses a model or reciprocal and checks by multiplication. → Accept either method and ask what the quotient counts.

Common error: Inverts the dividend. Repair: Check quotient × divisor = dividend.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

298. Week 4, Saturday
Watch and discuss

299. Today
Watch and discuss

300. Recall
Day 022 · Workbook p. 30 · R1, R2, R3, R4

301. Recall: answers
Day 022 · Workbook p. 30 · R1, R2, R3, R4

302. Whole number by unit fraction
Day 022 · Workbook p. 30 · P1

303. Whole number by non-unit fraction
Day 022 · Workbook p. 30 · P2, P3

304. The model
Watch and discuss

305. Fraction quotient is non-whole
Day 022 · Workbook p. 30 · P4, P5

306. Find amount for one whole group
Day 022 · Workbook p. 30 · P6, P7

307. Worked example · 1 of 3
Watch and discuss · worked example

308. Worked example · 2 of 3
Watch and discuss · worked example

309. Worked example · 3 of 3
Watch and discuss · worked example

310. Your turn
Day 022 · Workbook p. 30 · F

311. Two ways to see it
Watch and discuss · compare the same example

312. A claim to check
Day 022 · Workbook p. 30 · D

313. Words and meaning
Watch and discuss

314. Guided practice · X1
Day 022 · Workbook p. 30 · X1

315. Guided practice · X2
Day 022 · Workbook p. 30 · X2

316. Guided practice · X3
Day 022 · Workbook p. 30 · X3

317. A remainder has two possible units
Watch and discuss · draw on separate paper

318. A remainder has two possible units
Watch and discuss · draw on separate paper

319. Challenge C1
Day 022 · Workbook p. 30 · C1

320. Challenge C1 · Discuss
Day 022 · Workbook p. 30 · C1

321. Challenge C2
Day 022 · Workbook p. 30 · C2

322. Challenge C2 · Discuss
Day 022 · Workbook p. 30 · C2

323. In your own words
Day 022 · Workbook p. 30 · S

324. Exit ticket
Day 022 · Workbook p. 30 · E

325. The week ahead
Days 024-028 · Workbook pp. 33-37

326. Tutor reference
Tutor only;hidden in slide show

Sunday

327. Week 4, Sunday
Watch and discuss

328. Today
Watch and discuss

329. Recall
Day 023 · Workbook p. 31 · R1, R2, R3, R4

330. Recall: answers
Day 023 · Workbook p. 31 · R1, R2, R3, R4

331. Fraction divided by whole number
Day 023 · Workbook p. 31 · P1

332. Whole number divided by fraction
Day 023 · Workbook p. 31 · P2, P3

333. The model
Watch and discuss

334. Mixed number divided by fraction
Day 023 · Workbook p. 31 · P4, P5

335. Missing rectangle side
Day 023 · Workbook p. 31 · P6

336. Fractional-edge prism volume
Day 023 · Workbook p. 31 · P7

337. Worked example · 1 of 3
Watch and discuss · worked example

338. Worked example · 2 of 3
Watch and discuss · worked example

339. Worked example · 3 of 3
Watch and discuss · worked example

340. Your turn
Day 023 · Workbook p. 31 · F

341. Two ways to see it
Watch and discuss · compare the same example

342. A claim to check
Day 023 · Workbook p. 31 · D

343. Words and meaning
Watch and discuss

344. Mixed review
Day 023 · Workbook p. 31 · M1, M2, M3

345. A reciprocal counts groups in one whole
Watch and discuss · draw on separate paper

346. A reciprocal counts groups in one whole
Watch and discuss · draw on separate paper

347. Challenge C1
Day 023 · Workbook p. 31 · C1

348. Challenge C1 · Discuss
Day 023 · Workbook p. 31 · C1

349. Challenge C2
Day 023 · Workbook p. 31 · C2

350. Challenge C2 · Discuss
Day 023 · Workbook p. 31 · C2

351. In your own words
Day 023 · Workbook p. 31 · S

352. Exit ticket
Day 023 · Workbook p. 31 · E

353. The week ahead
Days 024-028 · Workbook pp. 33-37

354. Tutor reference
Tutor only;hidden in slide show

Answer review

355. Week 4 answers
Next lesson: slide 395. Use this review section after Friday.

356. The meaning of fraction division · 1 of 6
Day 022 · Workbook p. 30 · Items R1, R2, R3, R4

357. The meaning of fraction division · 2 of 6
Day 022 · Workbook p. 30 · Items P1, P2, P3, P4

358. The meaning of fraction division · 3 of 6
Day 022 · Workbook p. 30 · Items P5, P6, P7, F

359. The meaning of fraction division · 4 of 6
Day 022 · Workbook p. 30 · Items D, X1, X2

360. The meaning of fraction division · 5 of 6
Day 022 · Workbook p. 30 · Items X3, S, E, C1

361. The meaning of fraction division · 6 of 6
Day 022 · Workbook p. 30 · Items C2

362. Fraction division and reciprocals · 1 of 5
Day 023 · Workbook p. 31 · Items R1, R2, R3, R4

363. Fraction division and reciprocals · 2 of 5
Day 023 · Workbook p. 31 · Items P1, P2, P3, P4

364. Fraction division and reciprocals · 3 of 5
Day 023 · Workbook p. 31 · Items P5, P6, P7, F

365. Fraction division and reciprocals · 4 of 5
Day 023 · Workbook p. 31 · Items D, M1, M2, M3

366. Fraction division and reciprocals · 5 of 5
Day 023 · Workbook p. 31 · Items S, E, C1, C2

367. Dividing fractions · 1 of 6
Day 024 · Workbook p. 33 · Items 1, 2, 3, 4, 5

368. Dividing fractions · 2 of 6
Day 024 · Workbook p. 33 · Items 6, 7, 8, 9

369. Dividing fractions · 3 of 6
Day 024 · Workbook p. 33 · Items 10, 11, 12

370. Dividing fractions · 4 of 6
Day 024 · Workbook p. 33 · Items 13, 14, 15

371. Dividing fractions · 5 of 6
Day 024 · Workbook p. 33 · Items 16, 17, 18, C1

372. Dividing fractions · 6 of 6
Day 024 · Workbook p. 33 · Items C2

373. Dividing fractions · 1 of 5
Day 025 · Workbook p. 34 · Items 1, 2, 3, 4, 5

374. Dividing fractions · 2 of 5
Day 025 · Workbook p. 34 · Items 6, 7, 8, 9

375. Dividing fractions · 3 of 5
Day 025 · Workbook p. 34 · Items 10, 11, 12, 13

376. Dividing fractions · 4 of 5
Day 025 · Workbook p. 34 · Items 14, 15, 16, 17

377. Dividing fractions · 5 of 5
Day 025 · Workbook p. 34 · Items 18, C1, C2

378. Dividing fractions · 1 of 6
Day 026 · Workbook p. 35 · Items 1, 2, 3, 4, 5

379. Dividing fractions · 2 of 6
Day 026 · Workbook p. 35 · Items 6, 7, 8, 9

380. Dividing fractions · 3 of 6
Day 026 · Workbook p. 35 · Items 10, 11, 12, 13

381. Dividing fractions · 4 of 6
Day 026 · Workbook p. 35 · Items 14, 15, 16, 17

382. Dividing fractions · 5 of 6
Day 026 · Workbook p. 35 · Items 18, 19, 20, 21

383. Dividing fractions · 6 of 6
Day 026 · Workbook p. 35 · Items 22, 23, C1, C2

384. Dividing fractions · 1 of 5
Day 027 · Workbook p. 36 · Items 1, 2, 3, 4, 5

385. Dividing fractions · 2 of 5
Day 027 · Workbook p. 36 · Items 6, 7, 8, 9

386. Dividing fractions · 3 of 5
Day 027 · Workbook p. 36 · Items 10, 11, 12, 13, 14

387. Dividing fractions · 4 of 5
Day 027 · Workbook p. 36 · Items 15, 16, 17, 18

388. Dividing fractions · 5 of 5
Day 027 · Workbook p. 36 · Items C1, C2

389. Weekly check · 1 of 6
Day 028 · Workbook p. 37 · Items 1, 2, 3, 4

390. Weekly check · 2 of 6
Day 028 · Workbook p. 37 · Items 5, 6, 7, 8

391. Weekly check · 3 of 6
Day 028 · Workbook p. 37 · Items 9, 10, 11

392. Weekly check · 4 of 6
Day 028 · Workbook p. 37 · Items 12, 13, 14, 15

393. Weekly check · 5 of 6
Day 028 · Workbook p. 37 · Items 16, 17, 18, C1

394. Weekly check · 6 of 6
Day 028 · Workbook p. 37 · Items C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 022 · The meaning of fraction division

Workbook p. 30 · assigned set

R1 · F-COMP
$\frac{7}{8}$

Reason: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

R2 · R-EQUIV
20 blue tiles

R3 · R-RATE
0.65 dollars per marker

R4 · F-DIV
20; $\frac{1}{20}$

P1 · F-DIV
20 pieces

P2 · F-DIV
6 scoops

P3 · F-DIV
6 pieces

P4 · F-DIV
$1\frac{1}{4}$ groups

P5 · F-DIV
$\frac{2}{5}$ of a scoop

P6 · F-DIV
$\frac{9}{10}$ liter

P7 · F-DIV
$1\frac{1}{2}$ groups. Stories vary.

F · F-DIV
2 eighths; $2\frac{1}{2}$ groups

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · F-DIV
It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

Reason: He uses a familiar fraction rule without identifying the group. Rename one third as two sixths and compare with five sixths.

X1 · F-DIV1
$\frac{7}{2}$

X2 · F-DIV1
$\frac{20}{3}$ scoops

X3 · F-DIV1
$\frac{3}{28}$ liter

S · EXPLAIN
Answers vary. The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

Reason: Accept an accurate explanation supported by a matching example.

E · F-DIV
$\frac{8}{15}$ of the length

Reason: Use twentieths. The target length is 15 twentieths; the available amount is 8 twentieths. That is eight fifteenths of one group.

C1 · F-DIV1
3 full pieces; $\frac{1}{4}$ meter remains; quotient $\frac{10}{3}$ counts 3 and $\frac{1}{3}$ of a piece

Reason: Rename $2\frac{1}{2}$ as $\frac{10}{4}$ meters. Each piece uses 3 fourths, so 10 fourths contain 3 full groups with 1 fourth left. The leftover is $\frac{1}{3}$ of a $\frac{3}{4}$-meter piece, not $\frac{1}{3}$ meter.

C2 · F-DIV1
A: 2 cups; B: $\frac{12}{5}$ cups; both fill 2 full cups. B has $\frac{1}{6}$ liter left

Reason: A has 6 eighths, so it fills two 3-eighth cups. B has 12 twelfths, so it contains $\frac{12}{5}$ cupfuls: two full cups use $\frac{10}{12}$ liter, leaving $\frac{2}{12} = \frac{1}{6}$ liter.

Day 023 · Fraction division and reciprocals

Workbook p. 31 · assigned set

R1 · F-ADD
$\frac{5}{6}$

R2 · R-TABLE
12

R3 · P-PERCENT
75 %

R4 · F-DIV
12 pieces

P1 · F-DIV
$\frac{3}{20}$ liter

P2 · F-DIV
$6\frac{2}{3}$ groups

P3 · F-DIV
$3\frac{1}{2}$

Reason: One quarter is two eighths. Seven eighths contains seven halves of a two-eighths group.

P4 · F-DIV
3

P5 · F-DIV
4 pieces

P6 · F-DIV
$1\frac{1}{4}$ meters

P7 · G-VOL
1 cubic meter

F · F-DIV
$\frac{3}{2}$; $1\frac{1}{5}$

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · F-DIV
Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\frac{1}{2}$.

Reason: She reverses the dividend instead of changing the size of the divisor-sized group. Check whether $\frac{2}{7}$ groups of one fourth could make seven eighths.

M1 · F-MULT
2

M2 · R-EQUIV
3 cups

M3 · R-RATE
0.65 dollars per marker

S · EXPLAIN
Answers vary. Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

Reason: Accept an accurate explanation supported by a matching example.

E · F-DIV
7 pieces

Reason: $2\frac{5}{8} = \frac{21}{8}$. Twenty-one eighths make seven groups of three eighths.

C1 · F-DIV1
Length $\frac{5}{4}$ meters; perimeter $\frac{23}{6}$ meters

Reason: Length = area ÷ width = $(\frac{5}{6}) \div (\frac{2}{3}) = \frac{5}{4}$. Add length and width, then double: $2 \times (\frac{5}{4} + \frac{2}{3}) = 2 \times \frac{23}{12} = \frac{23}{6}$ meters. Check length × width gives the original area.

C2 · G-VOL
Each box: $\frac{5}{4}$ m³; unoccupied volume: $\frac{1}{2}$ m³

Reason: The base area is $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ m². Multiply by $\frac{5}{2}$ to get $\frac{5}{4}$ m³ per box. Two boxes use $\frac{5}{2}$ m³, leaving $3 - \frac{5}{2} = \frac{1}{2}$ m³.

Day 024 · Dividing fractions

Workbook p. 33 · assigned set

1 · N-FACT
18

2 · N-FACT
42

3 · N-FACT
24

4 · N-FACT
54

5 · P-PART
4, 20, and 30 counters

6 · F-DIV
20; $\frac{1}{20}$

7 · N-FACT
14

8 · F-DIV
$\frac{2}{3}$

9 · F-DIV · review
$\frac{5}{8}$

10 · F-DIV · review
$\frac{10}{3}$

11 · F-DIV · review
$\frac{5}{3}$

12 · F-DIV · review
$\frac{25}{3}$

13 · F-DIV
It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

14 · F-DIV1
$\frac{5}{4}$

15 · F-DIV1
$\frac{28}{3}$ scoops

16 · F-DIV1
$\frac{1}{10}$ liter

17 · F-MULT
$\frac{8}{5}$ kilograms

18 · F-MULT
$\frac{21}{40}$

C1 · F-DIV1
Capacity $\frac{9}{10}$ liter; extra $\frac{3}{10}$ liter

Reason: Two equal thirds of the jar hold $\frac{3}{5}$ liter, so one third holds $\frac{3}{10}$ liter. Three thirds hold $\frac{9}{10}$ liter. The missing third is $\frac{3}{10}$ liter.

C2 · F-DIV1
One batch: $\frac{16}{15}$ cups; $1\frac{1}{2}$ batches: $\frac{8}{5}$ cups

Reason: Find oil per full batch: $(\frac{4}{5}) \div (\frac{3}{4}) = \frac{16}{15}$. Then scale that amount by $\frac{3}{2}$ to get $\frac{8}{5}$ cups. As a check, $1\frac{1}{2}$ batches is twice the original $\frac{3}{4}$ batch, so oil doubles from $\frac{4}{5}$ to $\frac{8}{5}$.

Day 025 · Dividing fractions

Workbook p. 34 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · P-PART
42 points

6 · F-LINE
At $1\frac{3}{4}$, the seventh quarter-step from 0

7 · N-FACT
12

8 · F-DIV
$4\frac{2}{3}$

9 · F-DIV · review
$\frac{2}{5}$

Reason: Two and one half is five halves. Five halves times two fifths equals one.

10 · F-DIV · review
2

11 · F-DIV · review
$\frac{20}{9}$

12 · F-MULT · review
$\frac{16}{15}$

13 · F-DIV
Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\frac{1}{2}$.

14 · F-DIV1
$\frac{3}{2}$

15 · F-DIV1
$\frac{12}{5}$ scoops

16 · F-DIV1
$\frac{1}{16}$ liter

17 · F-DIV1
$\frac{95}{24}$

Reason: Rename the mixed number as $\frac{19}{6}$. Multiply by the reciprocal of $\frac{4}{5}$; then multiply the quotient by the original divisor.

18 · F-DIV1
1 meters

C1 · F-DIV1
8 complete meals; $\frac{3}{10}$ kilogram left
Reason: The exact quotient is $(\frac{7}{2}) \div (\frac{2}{5}) = \frac{35}{4} = 8\,\frac{3}{4}$ meals. Eight meals use $\frac{16}{5}$ kilograms. $\frac{7}{2} - \frac{16}{5} = \frac{35}{10} - \frac{32}{10} = \frac{3}{10}$ kilogram remains.

C2 · F-DIV1
First length $2\,\frac{1}{2}$ meters; second 3 meters. The narrower poster is longer
Reason: Divide the common area $\frac{15}{8}$ by each width. Results are $\frac{5}{2}$ and 3. To preserve the same area, a smaller positive width must pair with a larger length.

Day 026 · Dividing fractions

Workbook p. 35 · assigned set

1 · N-SCALE
62

2 · N-SCALE
6

3 · N-SCALE
13

4 · N-SCALE
18

5 · F-DIV
$4\,\frac{1}{2}$

6 · F-DIV
$\frac{5}{8}$

7 · R-TABLE
16

8 · R-EQUIV
5 cups

9 · F-ADD
$\frac{5}{6}$

10 · F-MULT
$\frac{3}{5}$

11 · N-FACT
8

12 · G-VOL
24 cm³

13 · PUZZLE · review
3 one-third-liter scoops and 2 half-liter scoops

14 · F-DIV1
$\frac{3}{2}$

15 · F-DIV1
4 scoops

16 · F-DIV1
$\frac{1}{7}$ liter

17 · F-DIV1
$\frac{25}{8}$
Reason: Rename the mixed number as $\frac{5}{4}$. Multiply by the reciprocal of $\frac{2}{5}$; then multiply the quotient by the original divisor.

18 · F-DIV1
$\frac{7}{4}$ meters

19 · G-VOL
4 cubic meters

20 · R-RATE
9 km per hour

21 · R-RATE
$0.70 per marker

22 · R-RATE
4 cards/min; $\frac{1}{4}$ min/card

23 · R-RATE
8 miles per hour

C1 · F-DIV1
$\frac{1}{8}$ liter per cup; five shares plus $\frac{1}{4}$ liter equal $\frac{7}{8}$ liter
Reason: The remainder is $\frac{7}{8} - \frac{2}{8} = \frac{5}{8}$ liter. Divide by 5 to get $\frac{1}{8}$ liter in each cup. Five eighths plus two eighths reconstruct seven eighths.

C2 · F-DIV1
One piece: $\frac{3}{10}$ meter; 2 meters holds $\frac{20}{3}$ piece-lengths and 6 complete pieces
Reason: Divide $\frac{4}{5}$ meter by $\frac{8}{3}$ pieces to get $\frac{3}{10}$ meter per piece. Divide 2 meters by $\frac{3}{10}$ meter per piece to get $\frac{20}{3}$ piece-lengths, or 6 and $\frac{2}{3}$. Thus 6 complete pieces fit; 7 would exceed 2 meters. Repeated multiplication checks both relations.

Day 027 · Dividing fractions

Workbook p. 36 · assigned set

1 · N-EST
600

2 · N-EST
7

3 · N-EST
About 100

4 · N-EST
About 20

5 · F-DIV
$4\frac{1}{2}$

6 · F-DIV
$\frac{5}{8}$

7 · F-DIV
$\frac{3}{4}$

8 · R-GRAPH
Yes
Reason: Every vertical value is twice the horizontal value.

9 · P-PART · review
1.8000 dollars

10 · P-WHOLE · review
60 shells

11 · P-PERCENT · review
75 %

12 · P-PART · review
10 stickers

13 · F-DIV
1 meters

14 · F-DIV
Keep the dividend $\frac{7}{8}$. Multiply by the reciprocal of the divisor: $\frac{7}{8} \times 4 = 3\frac{1}{2}$.

15 · F-DIV1
$\frac{7}{4}$

16 · F-DIV1
$\frac{24}{5}$ scoops

17 · F-DIV1
$\frac{1}{18}$ liter

18 · F-DIV1
$\frac{11}{3}$
Reason: Rename the mixed number as $\frac{11}{5}$. Multiply by the reciprocal of $\frac{3}{5}$; then multiply the quotient by the original divisor.

C1 · F-DIV1
6 bags; $\frac{3}{4}$ meter extra for a seventh bag
Reason: Usable cloth is $5 - \frac{1}{2} = \frac{9}{2}$ meters. Dividing by $\frac{3}{4}$ gives 6 exactly. No usable cloth remains, so a seventh bag needs the full $\frac{3}{4}$ meter. The reserved half meter stays set aside.

C2 · G-VOL
Height 3 meters; new volume $\frac{9}{16}$ cubic meter
Reason: The base area is $\frac{3}{8}$ square meter. Height = $(\frac{9}{8}) \div (\frac{3}{8}) =$ 3 meters. Half the height is $\frac{3}{2}$ meters; $\frac{3}{8} \times \frac{3}{2} = \frac{9}{16}$ cubic meter, half the original volume.

Day 028 · Weekly check

Workbook p. 37 · assigned set

1 · F-DIV
$4\frac{1}{2}$

2 · F-DIV
$\frac{5}{8}$

3 · F-DIV
$\frac{3}{4}$

4 · F-DIV
$1\frac{2}{3}$ meters

5 · R-RATE

0.65 dollars per marker

6 · P-PART

4, 20, and 30 counters

7 · R-TABLE

20

8 · F-ADD

$\frac{5}{6}$

9 · F-MULT

$\frac{3}{5}$

10 · F-DIV

It asks what fraction of a five-sixths group fits in one third. The answer is $\frac{2}{5}$.

11 · F-DIV1

$\frac{2}{3}$

12 · F-DIV1

$\frac{15}{2}$ scoops

13 · F-DIV1

$\frac{1}{5}$ liter

14 · F-DIV1

$\frac{95}{24}$

Reason: Rename the mixed number as $\frac{19}{6}$. Multiply by the reciprocal of $\frac{4}{5}$; then multiply the quotient by the original divisor.

15 · F-DIV1

$\frac{3}{2}$ meters

16 · G-VOL

$\frac{3}{2}$ cubic meters

17 · P-PART

25; whole = 100

18 · P-WHOLE

40

C1 · F-DIV1

A: $\frac{1}{6}$ liter; B: $\frac{3}{16}$ liter. B receives $\frac{1}{48}$ liter more per person

Reason: Divide each amount by its number of people. Compare $\frac{1}{6} = \frac{8}{48}$ with $\frac{3}{16} = \frac{9}{48}$. The smaller total can give the larger share because fewer people share it.

C2 · F-DIV1

The correct quotient is $\frac{5}{6}$; $\frac{8}{15} \times \frac{4}{5} = \frac{32}{75}$, not $\frac{2}{3}$. The dividend is smaller than one divisor-sized group

Reason: Division must undo multiplication by $\frac{4}{5}$. The claimed answer fails because $\frac{8}{15} \times \frac{4}{5} = \frac{32}{75}$. The correct calculation is $\frac{2}{3} \times \frac{5}{4} = \frac{5}{6}$; $\frac{5}{6} \times \frac{4}{5} = \frac{2}{3}$. Since $\frac{2}{3} < \frac{4}{5}$, the total contains less than one full group.